

ScrewSense

A backlit vision station that tracks every screw during laptop repair

ASUS UGen AI League Innovation Hackathon · Stage I Proposal
Theme: Workplace AI · Track: Battlefield Lightning (ASUS UGen300 / Hailo-10H)
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Contents

1 **Intros**
Context · Expected outcomes — p. 3–4

2 **ScrewSense knowledge**
Screw anatomy · Screw types · Features to classify — p. 5–7

3 **Feasibility & Results**
Feasibility map · Tools · Hardware · Stage-wise validation · Issues 1–3 · Dataset · Training · Next steps — p. 8–17

References & source code: p. 18 · Appendix: p. 19 onward

Context: Guanghua Digital Plaza repair shops

Taipei’s largest cluster of PC repair shops
Guanghua Digital Plaza hosts many small repair counters that open laptops all day — battery, SSD, fan and keyboard replacements.

What we saw on the bench
Screws taken out of a laptop are put down wherever there is space. When the technician reassembles the unit, they have to search for screws and often cannot tell which screw came from which hole, largely because they cannot remember the order in which the screws were removed.

No tool today remembers this for them.



Many small shops

Small quantities, diverse varieties, limited space, no fixed process



Screws scattered

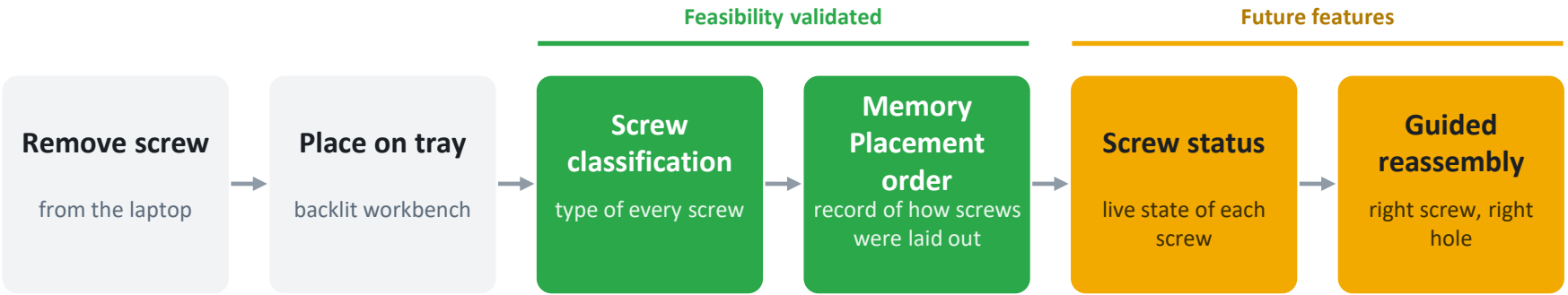
Different lengths and heads mixed on one bench



Slow reassembly

Time lost searching and trial-fitting screws

Expected outcomes: where ScrewSense is heading



Validated

Screw classification

Automatically identify each screw's type (diameter × length, head) as it is placed.

Validated

Placement-order record

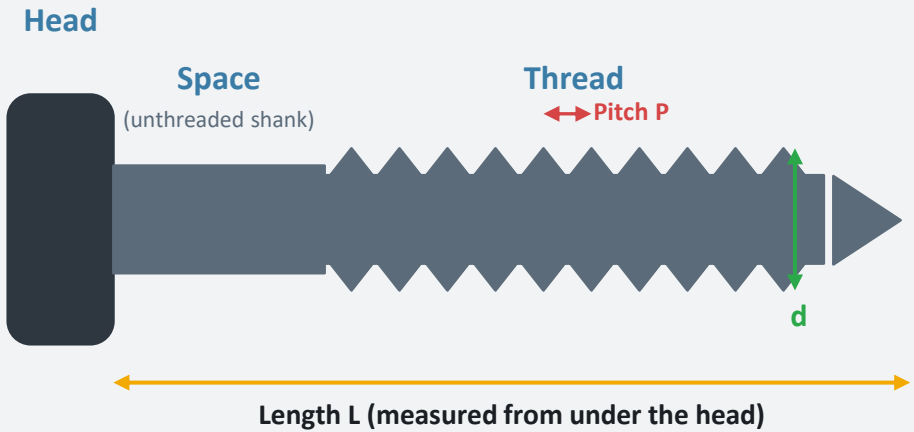
Log the order screws are put on the workbench, so reassembly can follow it in reverse.

Future

Screw status display

Show the state of every screw — on tray, reinstalled or missing.

Screw basics: anatomy of a screw



Head, Space and Thread are exactly the three classes our YOLOv8 model detects.

Head

Top of the screw; holds the drive (e.g. Phillips) and presses on the part. Pan, flat or wafer shapes.

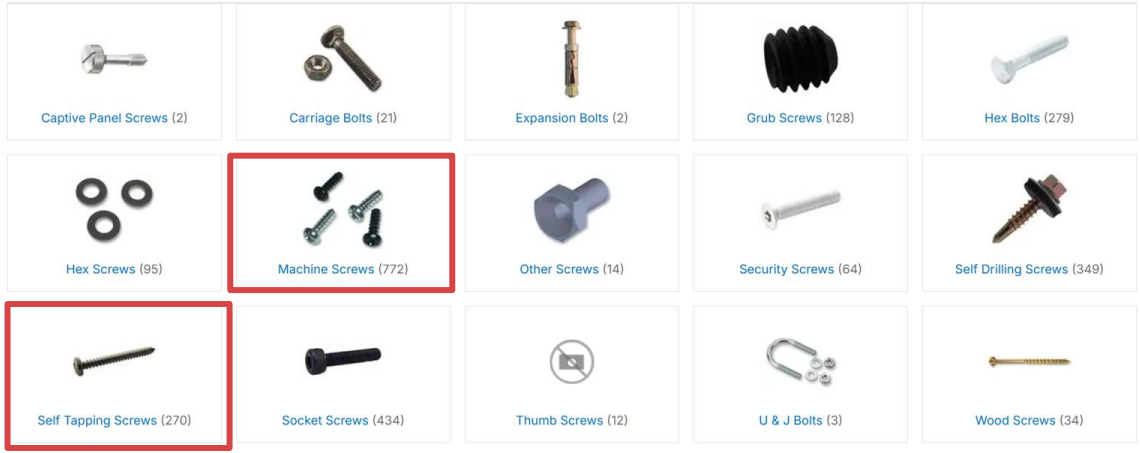
Space

Smooth, unthreaded section between head and thread; not every screw has one.

Thread

Helical ridge that grips the hole. Defined by diameter d and pitch P.

Screw types: machine vs self-tapping



Pointed tip **Blunt tip**

Red frames = the two types found in laptops. They are told apart mainly by the tip and thread.

Screws come in about 15 functional types. For laptop repair we only need to tell **two** apart:

1 Machine screws (our main)

Blunt tip

Used for most internal parts — CPU, GPU, fan…

2 Self-tapping screws (secondary)

Pointed tip

Used for the plastic housing.

Source: https://tw.element14.com/c/fasteners-mechanical/fasteners-fixings/screws-bolts

Screw features we need to classify

Detected today (YOLOv8) — Head | Thread | Space

To classify next

1. **Screw type**
   Machine (blunt tip) or self-tapping (pointed tip)
2. **Thread size & type**
   Diameter d and pitch P, e.g. M3 × 0.5
3. **Screw length**
   Length L, measured from under the head
4. **Head style**
   Head type: pan, flat, wafer…

M3 × 0.5 × 10

- **M** ISO metric thread
- **3** diameter d = 3 mm
- **0.5** pitch P = 0.5 mm
- **10** length L = 10 mm

Our scope

Now — feasibility stage
Common furniture sizes: larger and easier to see, to prove the method.

Future target — laptops
M2/M2.5 (highlighted): M2×3 vs M2×4 differ only in length; M2 vs M2.5 by 0.5 mm in diameter.

Feasibility map

Feasibility

Hardware → 3.3

1.	Camera	Verified	
2.	Base	Verified	product needs redesign
3.	Backlight	Verified	product needs redesign
4.	UI panel	In development	

Software → 3.4-3.9

1.	Screw parts	Verified	YOLOv8; product needs a new model
2.	Screw order	Verified	UI display to be built
3.	Screw types	In development	product needs more types
4.	UI design	In development	

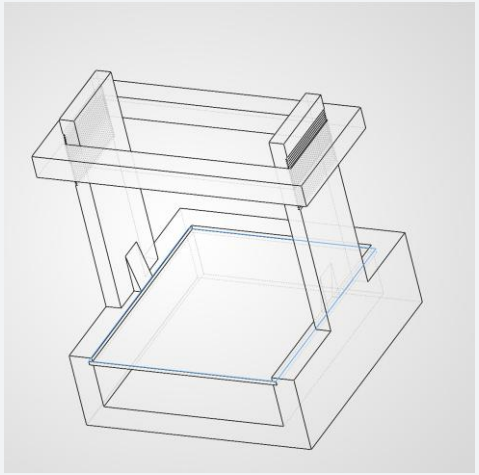
Tools, technologies & methodology

Software	What we use	Role
Detection model	Ultralytics YOLOv8n	Detects Head / Thread / Space in real time
Training data	Roboflow public screw dataset (CC BY 4.0)	1,195 labeled images, 3 classes
Model (new structure)	TransUNet	Extracts whole-screw masks while combining local CNN features with global Transformer context
Edge deployment	ONNX → Hailo HEF → ASUS UGen300	Evaluates on-device latency, accuracy, and operator compatibility

Hardware	What we use	Role
Design tool	SolidWorks	Design base, lens mount, shading baseplate
3D print	Our 3D printing material is PLA.	Printing temperature: 205°C, printing speed: 80 mm/s, infill density: 35%.
Camera lens	H703S Webcam	The H703S webcam will be fixed at a single angle to capture the image of screws on the shading baseplate and transmit the video feed to a computer.

Hardware

Design tool



SolidWorks — base, lens mount and shading baseplate

3D print



PLA · 205 °C · 80 mm/s · 35% infill

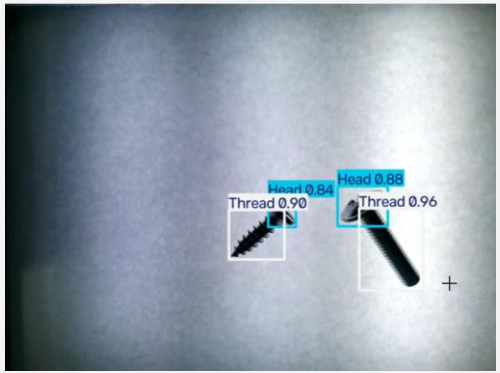
Camera lens



H703S webcam, fixed top-down above the backlit tray

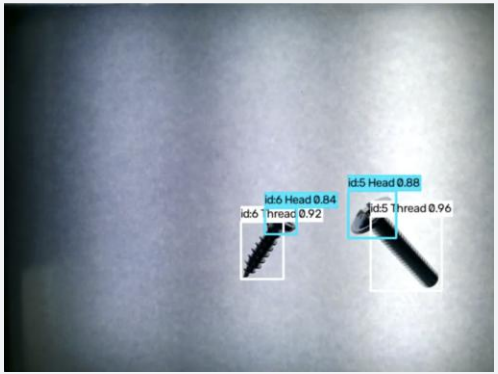
Stage-wise validation on our rig

Stage 1



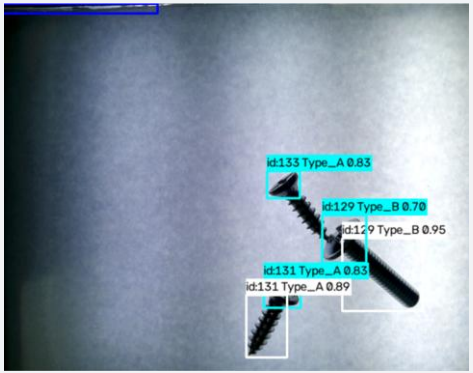
Recognise Head, Thread and Space and assign them to a type

Stage 2



Keep parts of different types from being mixed with each other

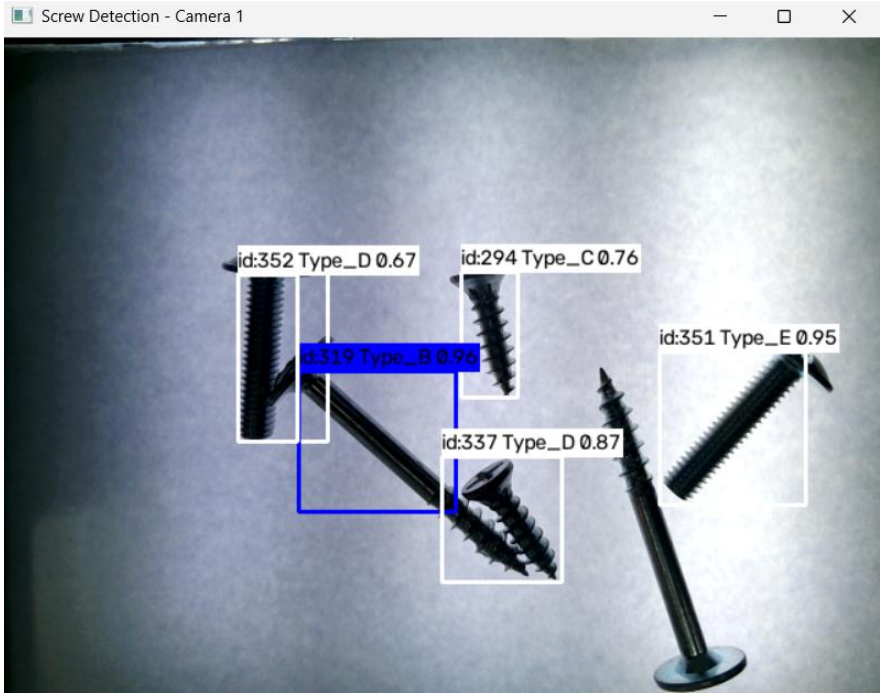
Stage 3



Recognise repeated screws of the same type

Screws are placed one by one on the backlit tray; each screw gets a stable ID and type.

Issue 1: overlapping screws reduce accuracy



Many screws added at once

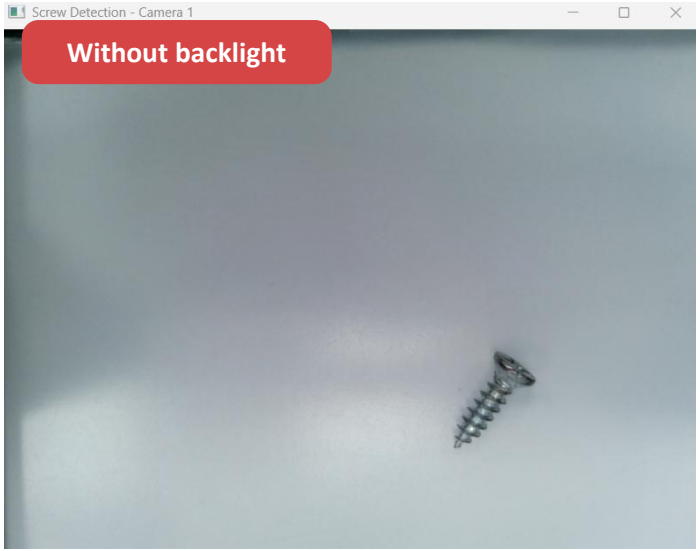
What happened

- id:337 Type_D 0.87 boxed the Thread of a different screw.
- One whole screw next to it was not detected at all.
- Crossing screws break the part-merging rule.

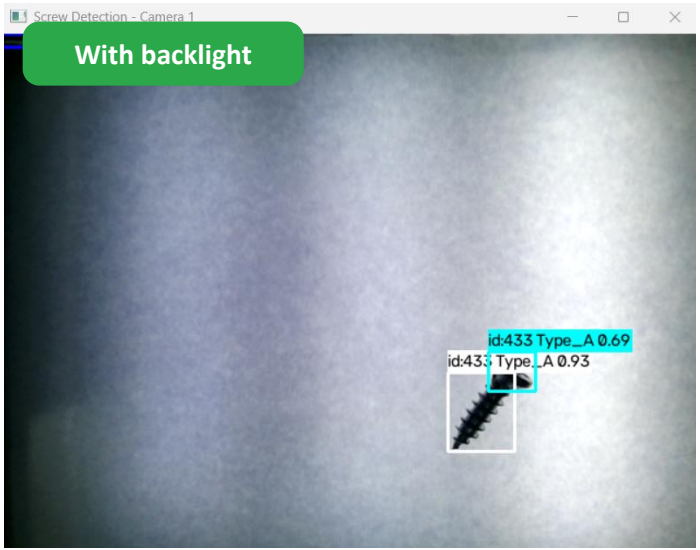
Our plan

- Optimise the attention / detection parameters
- Low-cost rule: flag blobs too large for one screw; split with watershed
- If still merged, ask the user to separate the screws

Issue 2: why the backlit panel matters



Blurry edges, nothing detected



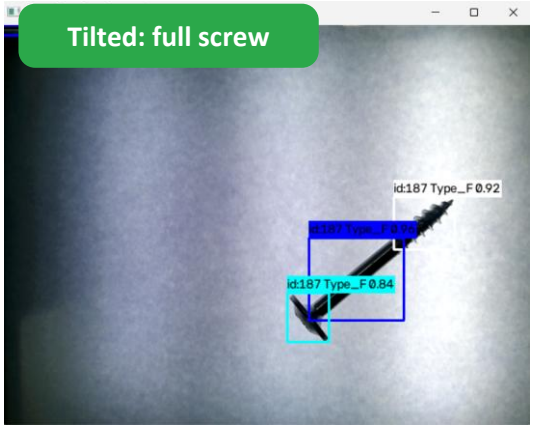
Sharp outline, full screw detected

Backlight makes clear silhouettes, so simple image processing can find screws without a heavy AI model.

With the backlit panel, the edges of the screw are clear and the system sees the complete outline, so it can judge the type. Without it, edges blend into the background and the system detects nothing.

Issue 3: same screw, different result







Cause A: uneven lighting

Three straight LED rows leave dark shadows in some areas, so the outline is incomplete.

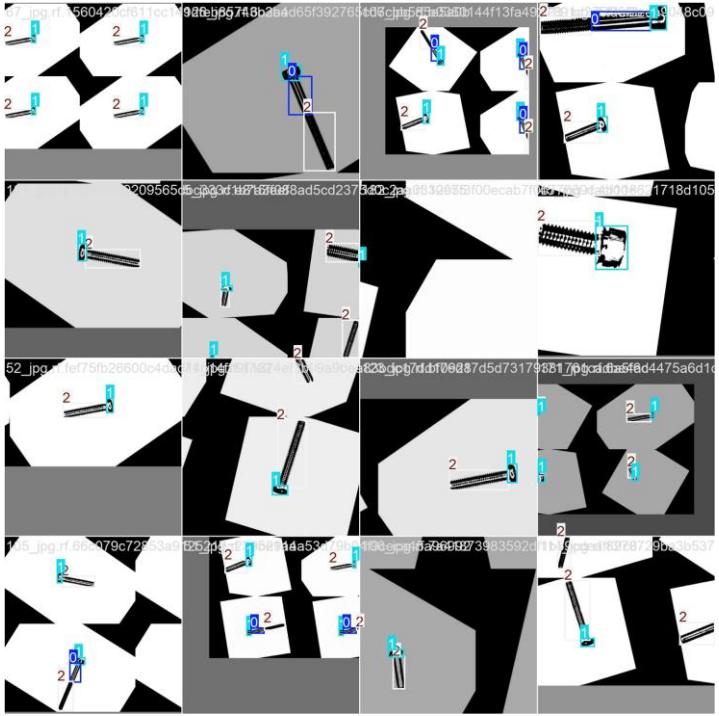
→ Matrix LED layout + side panel lights to remove shadows

Cause B: thin screw head

Seen straight from above, a thin head looks like part of the thread and is misread.

→ Measure length from the silhouette (geometry) + tune attention

Dataset: public screw images for the prototype



train_batch0 — augmented training images

1,195
labeled images (Roboflow screw-lwjvf)

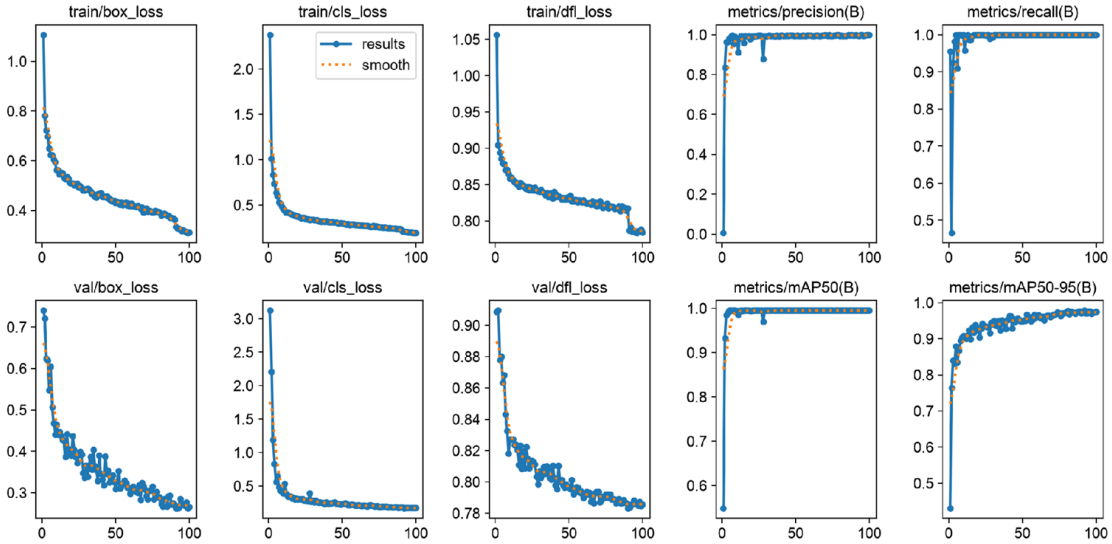
3
classes: Head · Thread · Space

×3
augmented versions: flips, ±45° rotation

Why a public dataset first? It let us validate the whole pipeline quickly. The screws in it are common hardware screws, larger than laptop screws, so we also tested with similar screws on our rig.

Next: our own dataset. Re-label the images with the features we must classify: screw type (machine / self-tapping), Thread size, Screw Length and Screw Head Style — starting with common laptop screws.

Training results



0.995	mAP50
0.974	mAP50-95
0.999	Precision
1.00	Recall

Results — loss and metric curves over 100 epochs

Reading the curves: train and val losses (box, cls, dfl) fall steadily with no sign of overfitting; precision, recall and mAP50 reach ~1.0 within ~10 epochs, while mAP50-95 keeps rising to 0.97, meaning boxes get tighter over time. *Note: these are validation-set scores for part detection; real-world accuracy on our rig is lower (see findings).*

Next steps: from prototype to product

Steps 1–3 showed that the current public dataset and YOLOv8 cannot reach the classification we need.
Steps 4–8 are our plan:

4	**Attention-based model**	Small objects are hard for YOLOv8. Use attention-based models (e.g. TransUNet) to classify pitch, thread type and head type on the public dataset.
5	**Own laptop-screw dataset**	Survey the screw models common in laptops and build our own dataset for the classification task.
6	**Auto Type A, B, C…**	Screws that cannot be classified are labelled Type A, B, C… automatically, in the order they are placed.
7	**Local web UI**	A web page on the local network shows the recognition results at a glance.
8	**Deploy on UGen300**	Deploy the whole pipeline on the ASUS UGen300 edge accelerator.

References & source code

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Chen, J. et al. TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation.arXiv:2102.04306, 2021.

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Fu, T., Chen, Y. SortScrews: A Dataset and Baseline for Real-time Screw Classification.

arXiv:2603.13027, 2026.

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Journal of Intelligent Manufacturing, 2026.

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Project repository

https://github.com/lhy4290/ScrewSense

Intro video

https://youtu.be/hLHHWtqhCiI